

Snyder Unit Hydrograph

See the USACE technical reference manual for a little recap of the method that was covered in earlier hydrology courses in the degree programme, and the basin lag equation:

<https://www.hec.usace.army.mil/confluence/hmsdocs/hmstrm/transform/snyder-unit-hydrograph-model>

Basin lag equation (standard lag) (equation 5 on the website listed above):

$$t_p = CC_t(LL_c)^{0.3}$$

where C_t = basin coefficient; L = length of the main stream from the outlet to the divide; L_c = length along the main stream from the outlet to a point nearest the watershed centroid; and C = a conversion constant (0.75 for SI and 1.00 for foot-pound system).